

Unit 8, Lesson 1: Volume of Right Rectangular Prisms



Benchmark

MA.6.GR.2.3 Solve mathematical and real-world problems involving the volume of right rectangular prisms with positive rational number edge lengths using a visual model and a formula.



Learning Target

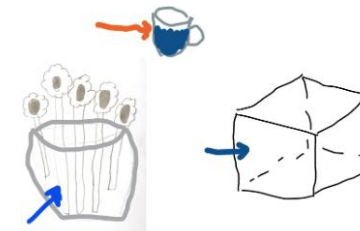
- ☐ I can pack a rectangular prism using a cube with fractional edges.



8.1.1 Warm-Up: Guess the Word

Micah and his family are playing a game in which each person chooses a card with a word written on it. Once a person chooses, they have one minute to draw a picture of the word. Then, his team guesses the word.

Micah chose the word “volume”. His drawing is shown.

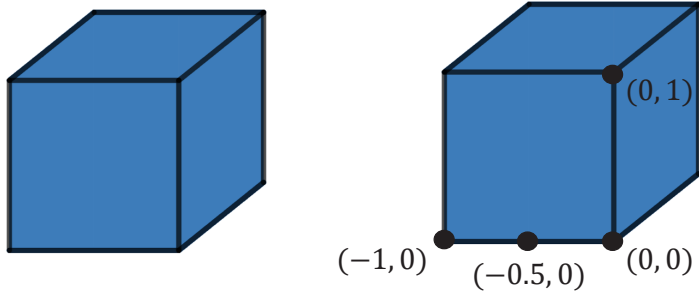


1. Choose the statement(s) that best describe volume. Select all that apply.
 - ☐ the sum of the length of all the edges of a three-dimensional object
 - ☐ the distance around a two-dimensional shape
 - ☐ the amount of three-dimensional space enclosed within a container
 - ☐ the total area of the surface of a three-dimensional object
 - ☐ the formula length times width times height



8.1.2 Exploration: The Unit Cube

- Two images of the same cube are shown. The image on the right shows the coordinates of some of the edges of the cube.



Share three things you know, two things you are not sure of, and one thing you have a question about.

I Know...	I'm Not Sure Of...	A Question I Have...

- This cube is known as a unit cube because each dimension (length, width, and height) is 1 unit.

What is the volume of the cube? Include the units for the cube's volume.



A **cube** is a rectangular prism with six congruent square faces.



8.1.3 Exploration: Drawing Fractional Cubes

1. Two cubes are shown.

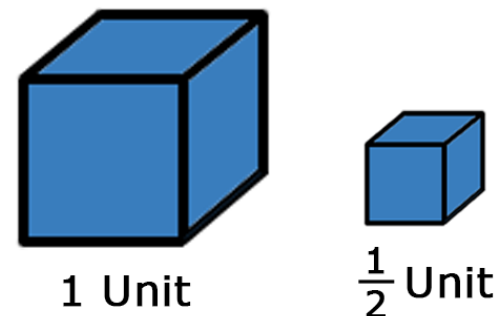


The cube on the left is a unit cube. The cube on the right has different dimensions.

Share an observation and something you wonder about this new cube.

Observation	Wondering

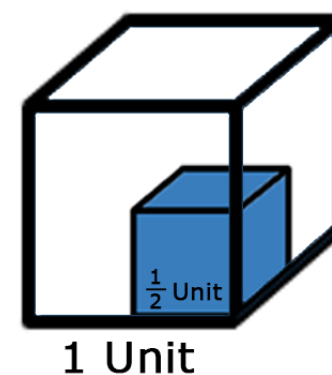
A unit cube and a half-unit cube are shown.



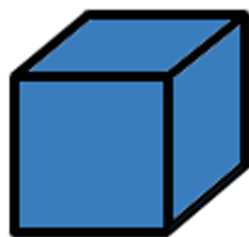
Make a guess of how many cubes with a length of $\frac{1}{2}$ unit, width of $\frac{1}{2}$ unit, and height of $\frac{1}{2}$ unit will fit in the 1-unit cube.

2. The figure shows a half-unit cube inside a unit cube.

Explain how the figure helps you to visualize how to pack cubes that have a fractional length.



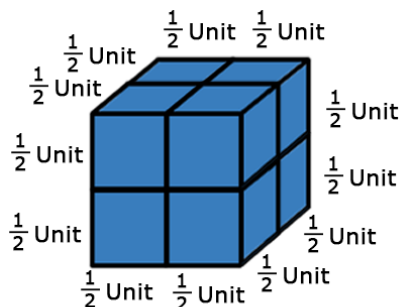
3. Three figures are shown.



1 Unit



$\frac{1}{2}$ Unit



a. Explain how the figures help you visualize how to pack cubes that have a fractional length.

b. To find the number of cubes whose dimensions are $\frac{1}{2}$ unit that fit into a cube whose dimensions are 1 unit, Derek wrote 1 as $\frac{2}{2}$ so he could see that $\frac{1}{2} + \frac{1}{2}$ would add to length of $\frac{2}{2}$ or 1. Complete his explanation.

Since $\frac{1}{2} + \frac{1}{2} = \frac{2}{2}$, or 1, I need _____ cubes for the length, _____ cubes for the width, and _____ cubes for the height.

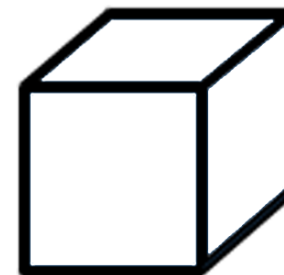
Since _____ $\times$ _____ $\times$ _____ = 8 I know that I need 8 cubes with dimensions $\frac{1}{2}$ unit to fit into one 1-unit cube.



8.1.4 Guided Instruction: Volume With Fractional Cubes

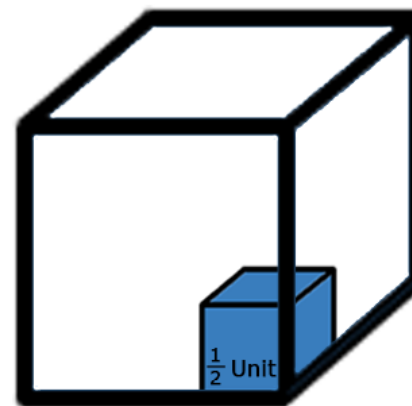
The cube shown is a scale model of a moving box.

1. The edges of the box are $1\frac{1}{2}$ feet long. How many cubes, whose dimensions are $\frac{1}{2}$ feet, can fit inside each dimension of the moving box?



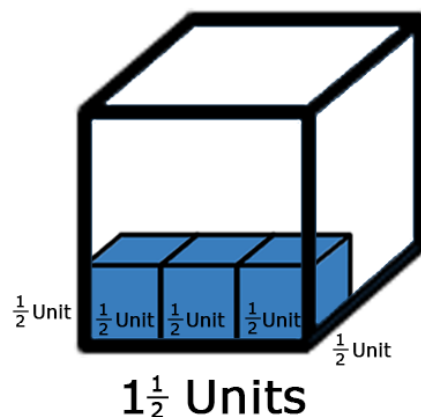
$1\frac{1}{2}$ Units

2. Sketch the cubes that can fit inside each dimension of the moving box.

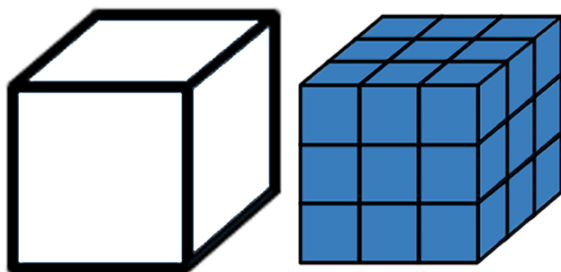


$1\frac{1}{2}$ Units

3. Since 3 cubes, whose dimensions are $\frac{1}{2}$ unit, will fit inside each dimension, how many of these cubes would be needed to pack the moving box?



4. This cube is a scale model of a moving box. The edges of the box are $1\frac{1}{2}$ feet long.



Since there are 27 cubes whose dimensions are $\frac{1}{2}$ feet that will fill the moving box, the volume of the moving box can be found by multiplying the volume of the fractional cube by 27. What is the volume of a cube whose dimensions are $\frac{1}{2}$ feet?

5. Since there are 27 cubes whose volume is $\frac{1}{8}$ cubic feet that will fill in the moving box, the volume of the moving box can be found by multiplying 27 and $\frac{1}{8}$ cubic feet. What is the volume of a cube of the moving box?

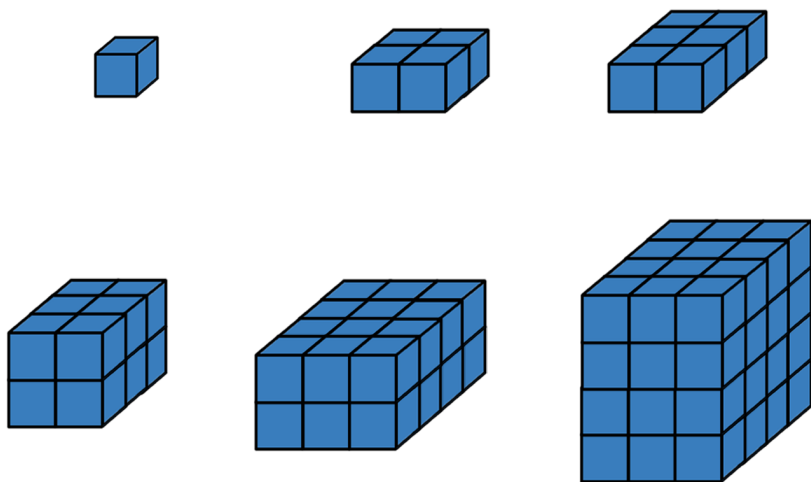
6. Find the volume of the moving box using the formula for volume of a rectangular prism or cube.

$$V = lwh$$

Each dimension of the box is $1\frac{1}{2}$ feet.

7. Does the volume you found using the formula match the volume you found by counting the number of fractional cubes?

8. The images show different rectangular prisms composed of various numbers of cubes.



a. What do you notice?

b. Explain how the dimensions of the prism can be determined using the cubes shown if each small blue cube is 1 unit cube.

9. The diagram shows 3 cubes whose dimensions are $\frac{1}{4}$ inches.



Select the statements that correctly describe the volume of the diagram as shown.

- ☐ Each cube has a volume of $\frac{1}{4}$ cubic inches.
- ☐ Each cube has a volume of $\frac{1}{16}$ cubic inches.
- ☐ Each cube has a volume of $\frac{1}{64}$ cubic inches.
- ☐ The volume of all three cubes is $\frac{3}{4}$ cubic inches.
- ☐ The volume of all three cubes is $\frac{3}{16}$ cubic inches.
- ☐ The volume of all three cubes is $\frac{3}{64}$ cubic inches.

10. The same diagram of 3 cubes whose dimensions are $\frac{1}{4}$ inches is shown.



- a. The dotted line represents the total length of the 3 cubes. How long, in inches, is the dotted line?

- (A) $\frac{4}{4}$ inches
- (B) $\frac{1}{4}$ inches
- (C) $\frac{3}{4}$ inches
- (D) $\frac{2}{4}$ inches

- b. The blue line represents the width of the 3 cubes.

What is the width, in inches?

- (A) $\frac{1}{4}$ inches
- (B) $\frac{2}{4}$ inches
- (C) $\frac{3}{4}$ inches
- (D) $\frac{4}{4}$ inches

- c. The green line represents the height of the cubes.

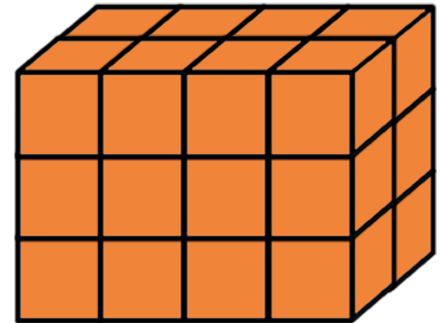
What is the height, in inches?

- (A) $\frac{1}{4}$ inches
- (B) $\frac{2}{4}$ inches
- (C) $\frac{3}{4}$ inches
- (D) $\frac{4}{4}$ inches

- d. Use the dimensions you found to determine the volume of all three cubes with the formula, $V = lwh$.

Finding volume using fractional unit cubes can be done by first finding the volume of one of the fractional unit cubes, and then multiplying that by the total number of fractional unit cubes.

11. A rectangular prism is shown.



- a. How many unit cubes whose dimensions are $\frac{1}{4}$ inch fill up the rectangular prism?

- b. Each unit cube whose dimensions are $\frac{1}{4}$ inch has a volume of $\frac{1}{64}$ cubic inches.

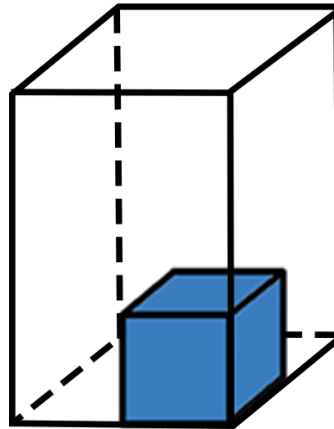
Find the total volume, in cubic inches, represented in the diagram by multiplying the number of cubes that fill the rectangular prism by the volume of one of the cubes.

- c. Find the total volume represented in the diagram, in cubic inches, using the formula, $V = lwh$. Show your work and your answer.



8.1.5 Your Turn!

- A rectangular prism with one cube inside is shown.
 - How many cubes will it take to fill the rectangular prism?



- The rectangular prism has a length of $\frac{16}{3}$ centimeters (cm), width of $\frac{16}{3}$ cm, and height of 8 cm as shown. Use the number of cubes in each dimension to determine the dimensions of the fractional unit cube.

- Complete the statement.

The unit cube shown has dimensions of $\frac{\square}{3}$ cm. It will take _____ of these unit cubes to fill the rectangular prism.

- Explain how to find the volume of the rectangular prism using the information from Part C.



8.1.6 Wrap-Up: Ticket Out the Door

- Match each image with its corresponding dimensions and volume. Use the table below to record your answers.

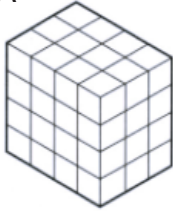
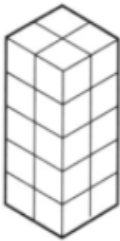
Image	Dimensions	Volume
A  The edges of each cube are $3\frac{1}{4}$ in. long.	D1 Length - 5 in. Width - 5 in. Height - 12.5 in.	V1 Volume $5\frac{1}{3}$ cubic in.
B  The edges of each cube are $\frac{2}{3}$ in. long.	D2 Length - 13 in. Width - 9.75 in. Height - 13 in.	V2 Volume 1,647.75 cubic in.
C  The edges of each cube are 2.5 in. long.	D3 Length - 2 in. Width - $(\frac{4}{3})$ in. Height - 2 in.	V3 Volume 312.5 cubic in.

Image	Dimensions	Volume
A		
B		
C		